PortPy Planning & Optimization for Radiation Therapy in Python

An Open-Source Platform for Benchmarking & Clinical Translation of Radiotherapy Treatment-Planning Optimization

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Project Goal

Goal: A Python-based, open-source project dedicated to radiotherapy **treatment-planning optimization.**

1

Benchmark Dataset

~200 lung & 130 prostate cases: CT, structures, IMRT/VMAT plans, and dose-influence matrices.

2

Baseline Algorithms

IMRT (fluence map, leaf sequencing) and VMAT (direct aperture) — AI and non-AI optimization.

3

Native Eclipse Integration

Produces Eclipse-compatible optimal fluence; uses Eclipse's beamlet dose calculation.

Three core features — each detailed in the next slides.

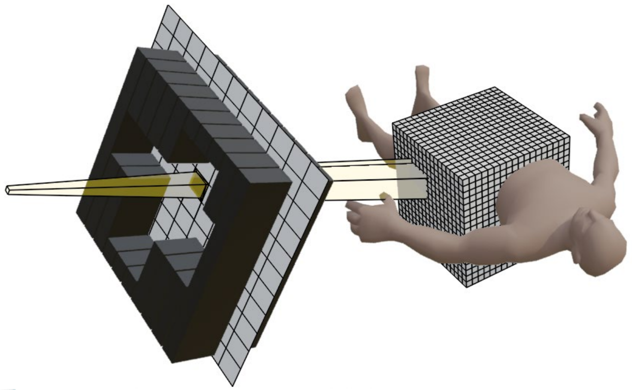
Benchmark Dataset

330 curated cases (CT, structures)
200 lung · 130 prostate
DICOM · TCIA

IMRT & VMAT plans
Reference ECHO plans
DICOM

Clinical criteria
e.g., lung mean-dose limits
JSON

Dose-influence matrix (72 beams)
Beamlet → voxel dose
JSON · HDF5



Pre-computed dose-influence matrix
- Beamlet-to-voxel dose for benchmark dataset

Want to use your own dataset?
Two ways:

- **Eclipse ESAPI script**
- **AI-based beamlet dose calculation**

Baseline Algorithms

1 **IMRT fluence-map optimization**
output → **Eclipse-compatible optimal fluence**

2 **IMRT fluence-map => Leaf sequencing**
output → **DICOM-RT**

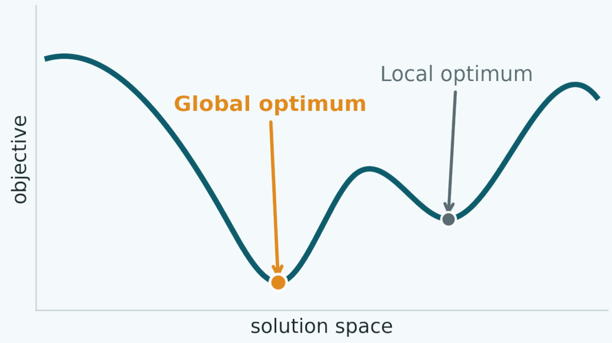
3 **VMAT aperture optimization**
output → **DICOM-RT**

4 **AI-based dose-prediction planning (IMRT & VMAT)**
AI predicted dose => IMRT/VMAT optimization => deliverable plan
output → **DICOM-RT**

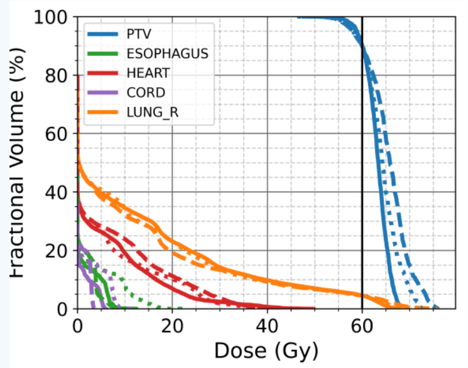
5 **Global optimization via Mixed Integer Programming**
Gold standard — but computationally expensive (~1 week)

All provided as ready-to-run **Jupyter notebooks.**

Global optimization — the gold standard



Non-convex problem — heuristics can stall in local optima.

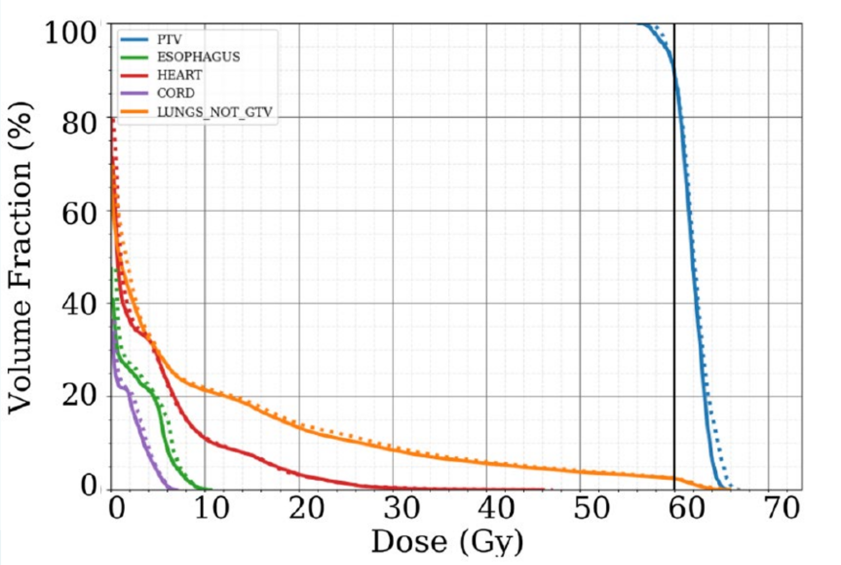


Plan quality (DVH) vs. global optimum *(~1 week to solve)*

Eclipse Integration — Clinical Deployment

PortPy plans are **dosimetrically accurate in Eclipse** — and **clinically deployed at MSK.**

✓ Dosimetric accuracy



Imported into Eclipse — **the Eclipse-recomputed dose matches the PortPy plan** with negligible discrepancy.

✓ Clinical deployment

Inside Eclipse	MSK Server	Inside Eclipse
1 Template plan & run ECHO	2 ECHO optimization	3 Final dose & plan review

Constrained hierarchical solve · ~30 min

12,000+ patients treated with ECHO at MSK

Coming Soon — GPU-Accelerated Optimization

Plug in any solver

PortPy works with a wide range of optimization engines out of the box:

OPEN-SOURCE

ECOS | OSQP | SCS

COMMERCIAL

MOSEK | GUROBI | CPLEX

Same code, any backend — switch solvers by changing a single argument.

COMING SOON

A GPU-accelerated engine, built for radiotherapy

~5x

faster than the best commercial solver

Purpose-built for the structure of radiotherapy optimization problems.

Summary

Powerful, open-source research tool

Fully customizable IMRT/VMAT optimization engine for radiotherapy.

Benchmark dataset

~200 lung and 130 prostate curated cases.

Baseline algorithms

IMRT (fluence map, leaf sequencing), VMAT (aperture), and global optimization.

Eclipse integration — clinically deployed

12,000+ patients treated with ECHO at MSK.

GPU-accelerated optimization coming soon

~5× faster than the best commercial solver.

40,000+ downloads (~1,000/month)

github.com/PortPy-Project

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